Dexmedetomidine

Authors

Bradley Reel[1]; Christopher V. Maani[2].

Affiliations

[1] Uniformed Services University of Health
[2] US Army

Last Update: May 8, 2022.

Continuing Education Activity

The FDA-approved indications for dexmedetomidine are sedation of intubated and mechanically ventilated patients in the intensive care unit (ICU) and peri-procedural (or peri-operative) sedation of non-intubated patients. Over time, usage has expanded to off-label uses, including treatment and prevention of delirium, adjunctive analgesia, therapy for insomnia in the ICU, and treatment of alcohol withdrawal. Dexmedetomidine has also been used in peripheral nerve blocks to prolong the duration of analgesia. This activity will highlight the mechanism of action, adverse event profile, pharmacology, monitoring, and relevant interactions of dexmedetomidine, pertinent for members of the interprofessional team in treating patients where anesthesia is appropriate.

Objectives:

- Identify the mechanism of action of dexmedetomidine.
- Summarize the indicated uses for dexmedetomidine.
- Review the adverse effect profile and contraindications of dexmedetomidine.
- Explain the importance of improving care coordination among the interprofessional team to enhance care delivery for patients when using dexmedetomidine as an anesthesia agent.

Access free multiple choice questions on this topic.

Indications

The FDA-approved indications for dexmedetomidine are sedation of intubated and mechanically ventilated patients in the intensive care unit (ICU) and peri-procedural (or peri-operative) sedation of non-intubated patients. Over time, usage has expanded to off-label uses, including treatment and prevention of delirium, adjunctive analgesia, therapy for insomnia in the ICU, and treatment of alcohol withdrawal. This expanded list of indications is because it commonly produces a sedated state allowing patients to be comfortable and cooperative during mechanical ventilation. Additionally, it does not require stoppage to accomplish extubation and is safe to use in non-intubated patients because it does not produce significant respiratory depression. Given THAT inherent analgesic properties are rare with most sedatives, dexmedetomidine produces an opioid-sparing effect.[1]

ICU sedation with dexmedetomidine has been demonstrated to decrease the incidence and duration of delirium and delay the onset of delirium compared with other sedatives.[2] Evidence also exists that there is a reduction in both the time to extubation and the number of ventilator-dependent hours in patients with delirium.[3] These reductions may be related to the ability of dexmedetomidine to reduce the need for other medications (for example, propofol, benzodiazepines, opioids) to promote a comfortable/cooperative patient. This effect has been particularly useful in the treatment of elderly post-cardiac surgery patients.[4] The beneficial effects in this particular patient population extend into longer outcomes, including survival at two years and improved cognitive function and quality of life. It has also become common to treat poor sleep in the ICU with dexmedetomidine infusions at doses as high as 1.5 mcg/kg per hr. This practice is supported by research showing dexmedetomidine often yields sleep quality closely related to stage 2 non-REM sleep.[5] However, it is also important to note that there is data showing alteration of normal sleep patterns, and subjects achieve no restorative rapid-eye-movement or slow-wave sleep. Additionally, studies demonstrating the benefit in clinical outcomes related to improved sleep generated by dexmedetomidine are sparse.[6]

Dexmedetomidine is commonly used in anesthesia practice as well. It is used for procedural sedation for a variety of operations. It is also used frequently for sedation during the performance of awake intubation. It is ideally suited for this indication for the reasons stated above. Dexmedetomidine is also an adjunct infusion during general anesthesia. There is evidence that dexmedetomidine decreases postoperative pain, postoperative opioid usage, and nausea.[7] This effect has also been shown to be present when using dexmedetomidine for sedation during procedures performed under spinal anesthesia.[8] However, at least one recent study has been called into question this opioid-sparing effect for a cohort of patients undergoing major spine surgery.[9] There has been some interest in using dexmedetomidine as an adjunct to prevent emergence agitation, postoperative delirium, and postoperative cognitive dysfunction. There is evidence for the prevention of emergence agitation in both children and adults.[10] [11]Because of enthusiasm over apparent prevention/treatment of delirium with dexmedetomidine when used for ICU sedation, it was hoped this benefit would be reproduced for postoperative delirium when using an intraoperative infusion of dexmedetomidine; however, a recent randomized trial did not show a statistically significant benefit in this postoperative population.[12] Dexmedetomidine has also been used in peripheral nerve blocks to prolong the duration of analgesia. Research suggests the dexmedetomidine may extend the duration of a peripheral nerve block for approximately 3 hours.[13]

Mechanism of Action

Dexmedetomidine is an alpha agonist having sedative, anxiolytic, hypnotic, analgesic, and sympatholytic properties. It produces these effects by inhibiting central sympathetic outflow by blocking the alpha receptors in the brainstem, thereby inhibiting the release of norepinephrine. It has a selectivity of 1600 to 1 for the alpha2 receptor as compared to alpha1. This selectivity is especially significant compared to another alpha agonist, clonidine, with a selectivity of 220 to 1. The mechanism by which dexmedetomidine may increase the duration of a peripheral nerve block is not fully understood but is believed to most likely be a perineural mechanism rather than a systemic or central mechanism, which appears to prolong the duration by blocking the cation current.[14]

Administration

For ICU sedation, the typical dosage range is 0.2 to 0.7 mcg/kg per hour. However, the dose can increase to 1.5 mcg/kg per hour to achieve the desired sedation level. Doses as high as 2.5 mcg/kg per hour have been described, but it is unlikely that doses higher than 1.5 mcg/kg per hour achieve any further therapeutic benefit in the absence of increased side effects. Dosage adjustments for renal or hepatic impairment are not required but should be considered, especially for hepatic impairment. The manufacturer does not recommend durations longer than 24 hours; however, longer durations have been demonstrated as safe and effective. Clinicians can administer a loading dose of 0.5 to 1.0 mcg/kg; this is usually avoidable if the medication is for labile critically ill patients or those with tenuous hemodynamics.

When used in anesthesia, the typical dosing is a loading dose of 0.5 to 1.0 mcg/kg, usually followed by a continuous infusion of 0.2 to 0.7 mcg/kg per hour titrated to desired sedation goals. As mentioned, higher infusion doses can help achieve the desired effect.

When used as an adjunct for peripheral nerve block, the dose of dexmedetomidine is usually 1 mcg/kg to achieve the desired prolongation.[15]

Adverse Effects

The most common adverse effects of dexmedetomidine are hypotension, bradycardia, and hypertension. Hypertension can result from the stimulation of alpha subtypes of receptors in vascular smooth muscles. Hypertension usually does not require treatment and can be avoided by the slow administration or omission of the loading dose. Hypotension and bradycardia are the results of the stimulation of presynaptic alpha receptors, which leads to a decreased release of norepinephrine; this is in addition to the decrease in the central sympathetic outflow. These are concerns regardless of the route of administration.[16]

Contraindications

There are no absolute contraindications to the use of dexmedetomidine. However, it should be used cautiously in patients with bradycardia and hypotension as the medication may exacerbate these findings. Additionally, it should be used cautiously in patients with known heart failure as there is level B evidence showing dexmedetomidine can potentially exacerbate myocardial dysfunction.

Monitoring

There are no specific drug monitoring requirements. The level of sedation, heart rate/rhythm, blood pressure, and pulse oximetry requires close monitoring.

Toxicity

At present, there is no chemical reversal or antidote for dexmedetomidine. Supportive care and close monitoring are the staples of treatment for overdose.

Enhancing Healthcare Team Outcomes

Dexmedetomidine is a useful non-opioid drug used to manage pain and sedation in the ICU and operating room. The drug is effective, but its use does require monitoring of vital signs. Both the pharmacist and clinician should ensure the administration of the right dose to prevent adverse effects. ICU nurses should know that the drug requires caution in patients with heart failure. All these disciplines working collaboratively as an interprofessional healthcare team will optimize outcomes when using dexmedetomidine. [Level 5] The primary downside to dexmedetomidine is its cost.